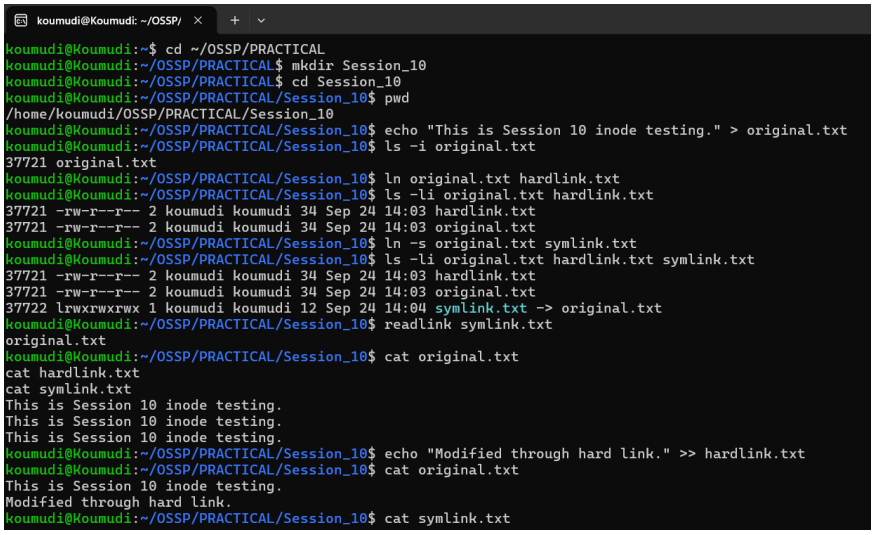
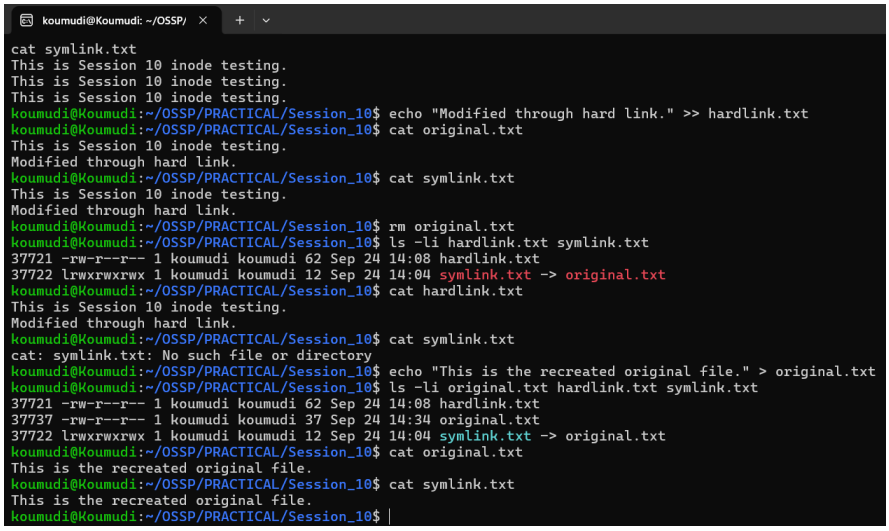
Practical (Week 10)

Session 10

Task1: Investigate inode structures using commands such as ls -i, stat, and find. Create hard links and symbolic links and explain their impact on inode allocation.





Task2: Design a program using mmap() to perform file reading and writing. Compare memory-mapped 1/0 with traditional read/write operations in terms of efficiency and Implementation complexity.

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2

```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    int fd;
    struct stat st;
    char *mapped;

    fd = open("mmap_input.txt", O_RDWR);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    mapped = mmap(NULL, st.st_size,
                  PROT_READ | PROT_WRITE,
                  MAP_SHARED, fd, 0);

    if (mapped == MAP_FAILED)
    {
        perror("mmap");
```

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2

```c
    mapped = mmap(NULL, st.st_size,
                  PROT_READ | PROT_WRITE,
                  MAP_SHARED, fd, 0);

    if (mapped == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original file content:\n");
    printf("%.*s\n", (int)st.st_size, mapped);

    const char *message = "Modified using mmap.";
    size_t message_length = strlen(message);

    if (message_length <= (size_t)st.st_size)
    {
        memcpy(mapped, message, message_length);
    }

    if (msync(mapped, st.st_size, MS_SYNC) == -1)
    {
        perror("msync");
    }

    printf("File modified successfully using mmap().\n");

    munmap(mapped, st.st_size);
    close(fd);

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ echo "This file will be modified using mmap." > mmap_input.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ cat mmap_input.txt
This file will be modified using mmap.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ nano mmap_file.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ gcc mmap_file.c -o mmap_file
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ ./mmap_file
Original file content:
This file will be modified using mmap.

File modified successfully using mmap().
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ cat mmap_input.txt
Modified using mmap.dified using mmap.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$
```

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2

```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main()
{
    int fd;
    ssize_t bytes_read;
    char buffer[BUFFER_SIZE];

    fd = open("mmap_input.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    bytes_read = read(fd, buffer, BUFFER_SIZE - 1);

    if (bytes_read == -1)
    {
        perror("read");
        close(fd);
        return 1;
    }

    buffer[bytes_read] = '\0';

    printf("File content using read():\n");
    printf("%s\n", buffer);

    close(fd);

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ nano traditional_io.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ gcc traditional_io.c -o traditional_io
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ ./traditional_io
File content using read():
Modified using mmap.dified using mmap.

koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ ^C
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ wc -l mmap_file.c traditional_io.c
  63 mmap_file.c
  39 traditional_io.c
 102 total
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$ ls -lh mmap_file traditional_io
-rwxr-xr-x 1 koumudi koumudi 17K Sep 24 14:39 mmap_file
-rwxr-xr-x 1 koumudi koumudi 16K Sep 24 14:42 traditional_io
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_10$
```